UML Diagram BUS RESERVATION SYSTEM

The Bus Reservation System is a console-based Java application that allows users to:

- View available buses
- Book seats in a bus
- View booking details

This system is implemented using Object-Oriented Programming concepts in Java and runs in a command-line environment.

The project demonstrates:

- Encapsulation
- Class relationships
- List collections (ArrayList)
- Conditional validation
- Menu-driven programming

This is a basic academic-level project without database or GUI integration.

UML Use Case Diagram for Bus Reservation System

Actors:

- User
  (Passenger)
- System

Use Cases:

- View Buses
  - User can see bus details (Bus No, AC type, Capacity)
- Book Ticket
  - User selects bus number
  - User selects seat number
  - User enters passenger name
  - System validates availability
- View Bookings
  - User can see all booking details
- Exit Application

- **Relationships**
  - User →
    - View Buses
    - Book Ticket
    - View Bookings
    - Exit
  - System →
    - Validate Seat Availability
    - Store Booking Data

Classes and Attributes

1. Bus

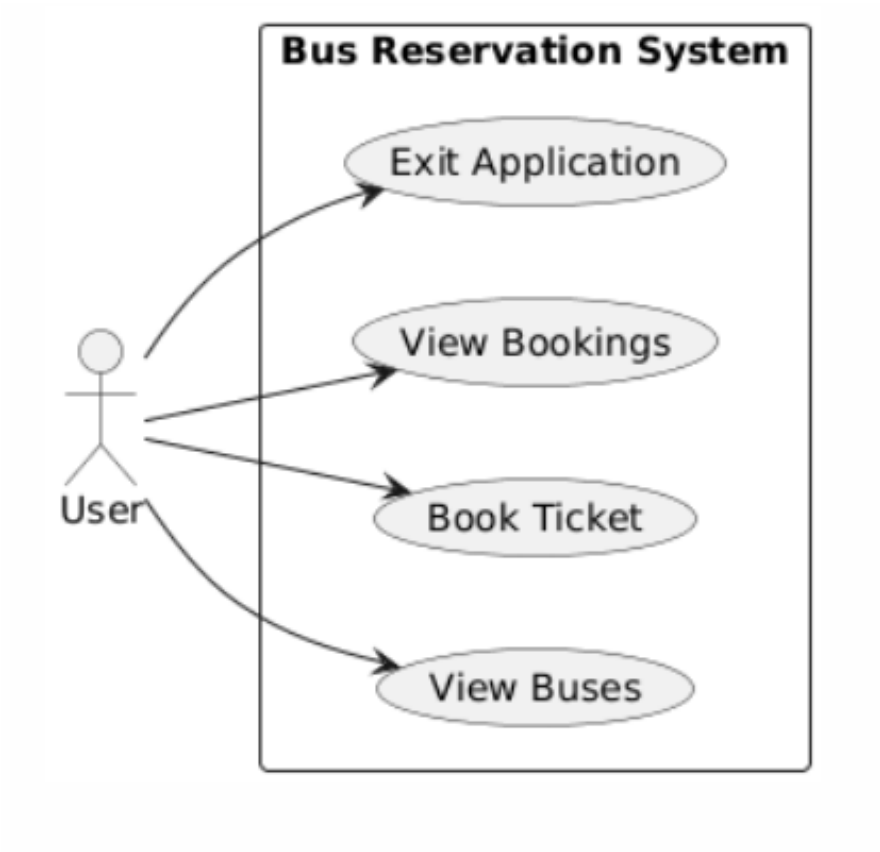
- Attributes:
  - busNo : int
  - ac : boolean
  - capacity : int
  - bookedSeats : List<Integer>
- Methods:
  - getBusNo()
  - isAc()
  - getCapacity()
  - isSeatAvailable(int seatNo)
  - bookSeat(int seatNo)
  - displayBusInfo()

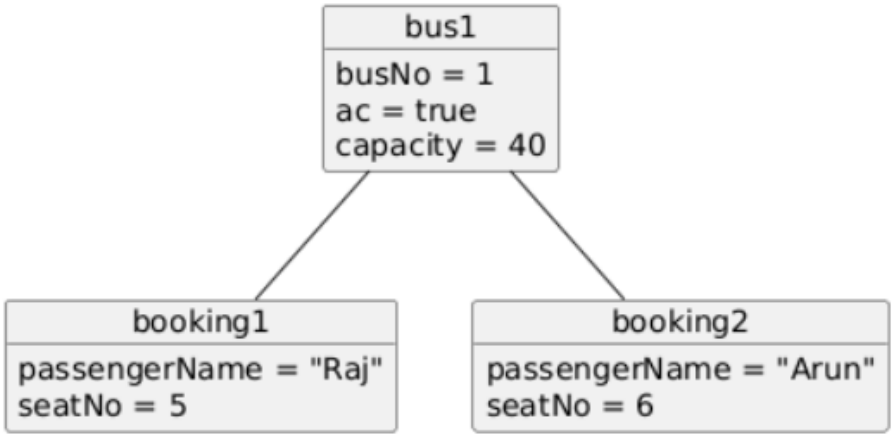
2. Booking

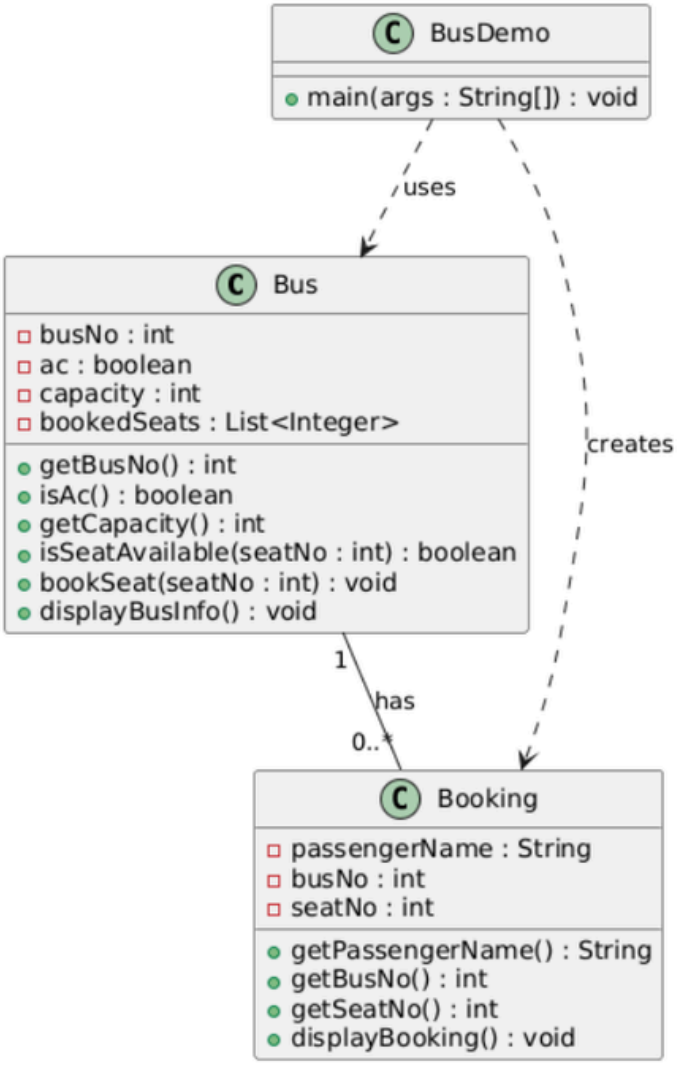
- Attributes:
  - passengerName : String
  - busNo : int
  - seatNo : int
- Methods:
  - getPassengerName()
  - getBusNo()

- getSeatNo()
- displayBooking()

3. BusDemo (Main Class)
    - Responsibilities:
    - Display menu
    - Handle user input
    - Manage bus list
    - Manage booking list
    - Control application flow




bus1
busNo = 1
ac = true
capacity = 40
booking1
passengerName = "Raj"
seatNo = 5
booking2
passengerName = "Arun"
seatNo = 6


BusDemo
main(args : String[]) : void
uses
creates
Bus
busNo : int
ac : boolean
capacity : int
bookedSeats : List<Integer>
getBusNo() : int
isAc() : boolean
getCapacity() : int
isSeatAvailable(seatNo : int) : boolean
bookSeat(seatNo : int) : void
displayBusInfo() : void
1
has
0..*
Booking
passengerName : String
busNo : int
seatNo : int
getPassengerName() : String
getBusNo() : int
getSeatNo() : int
displayBooking() : void